

## **Data Communication and Computer Networks Review Questions**

1. Which layer uses port number to identify applications?
  - A. Transport layer
  - B. Physical layer
  - C. Network layer
  - D. Application layer
2. Which one of the following is not correct about packet switching and circuit switching?
  - A. In packet switching messages are sent in small blocks.
  - B. In packet switching end to end connection has to be established
  - C. In circuit switching a channel is dedicatedly used
  - D. Packet switching is more efficient than circuit switching
3. Which network type is the largest as compared to the rest?
  - A. Wide Area Network
  - B. Metropolitan Area Network
  - C. The Internet
  - D. Local Area Network
4. Which one of the following is not correct about POP3 and SMTP?
  - A. A sender can send email using SMTP
  - B. Both are application layer protocols
  - C. A sender can receive emails using POP3
  - D. Both are network layer protocols

5. A university owns a number of private local area and wide area networks, which are designed for an access by its academic staff, students and administrative workers. Which term best describes the University's network?

A. Local Area Network

B. The Internet

C. Wide Area Network

D. Metropolitan Area Network

6. Which one of the following is different from the other based on its role in computer network?

A. Routers

B. Network printer

C. Wireless access point

D. Hubs

7. Suppose you are administering a network and would like to know whether a computer in the upstairs is connected to the network or not. Therefore, which command would you use?

A) nslookup

B) ping

C) DHCP discover

D) traceroute

8. At which layer of the OSI model devices such as switches and network interface cards (NIC) are used

A) Physical layer

B) Application Layer

C) Network Layer

D) Data link

layer

9. Which layer is responsible for logical addressing and routing?

A. Transport layer

B. Network layer

C. Data link layer

D. Application layer

10. What is the purpose of a MAC (Media Access Control) address?
- A. Identifying a device on a local network
  - B. Identifying a device on the internet
  - C. Providing encryption for data transmission
  - D. Resolving domain names to IP addresses
11. What is the purpose of DNS (Domain Name System) in computer networks?
- A. Assigning IP addresses to devices
  - B. Resolving domain names to IP addresses
  - C. Providing secure communication
  - D. Managing network traffic
12. What is the primary function of a router in a computer network?
- A. Connecting devices within the same network
  - B. Filtering and forwarding data between different networks
  - C. Providing wireless access to devices
  - D. Managing network security
13. What is the role of a firewall in network security?
- A. Filtering and monitoring network traffic
  - B. Providing physical security for network devices
  - C. Managing IP addresses in a network
  - D. Establishing secure connections between devices
14. Which networking device operates at the Data Link layer of the OSI model and filters traffic based on MAC addresses?
- A. Router
  - B. Switch
  - C. Hub

D. Bridge

15. What is a VLAN (Virtual Local Area Network) used for in networking?

- A. Connecting devices in different physical locations
- B. Creating logically segmented networks within a physical network
- C. Providing wireless access to devices
- D. Assigning IP addresses to devices

16. What is the purpose of NAT (Network Address Translation) in networking?

- A. Assigning public IP addresses to devices
- B. Translating private IP addresses to public IP addresses
- C. Filtering network traffic based on MAC addresses
- D. Establishing secure connections between devices

17. What is the function of a proxy server in a network?

- A. Filtering and monitoring web traffic
- B. Translating IP addresses to domain names
- C. Managing IP addresses in a network
- D. Establishing secure connections between devices

18. What is the primary advantage of using IPv6 (Internet Protocol version 6) over IPv4?

- A. Larger address space
- B. Faster data transfer speed
- C. Better security features
- D. Compatibility with older networking devices

19. What is the purpose of ICMP (Internet Control Message Protocol) in networking?

- A. Encrypting data for secure transmission
- B. Diagnosing network issues and errors
- C. Assigning IP addresses to devices
- D. Filtering web traffic based on content

20. What is the purpose of a subnet mask in IP networking?

- A. Identifying a device on a local network
- B. Dividing an IP address into network and host portions
- C. Resolving domain names to IP addresses
- D. Assigning a unique identifier to each device

21. In networking, what does the term "bandwidth" refer to?

- A. The physical size of a network
- B. The capacity of a network to transmit data
- C. The speed of data transfer in a network
- D. The number of devices in a network

22. What is the primary purpose of the Data Link layer in the OSI model?

- A. Establishing end-to-end communication between devices
- B. Providing logical addressing for devices
- C. Ensuring reliable data transfer between adjacent network nodes
- D. Managing routing and forwarding of data between networks

23. In a network, what is the role of a default gateway?

- A. Assigning IP addresses to devices
- B. Filtering network traffic based on MAC addresses
- C. Providing a path for data to exit the local network
- D. Resolving domain names to IP addresses

24. In networking, what does the term "latency" refer to?

- A. The physical size of a network
- B. The speed of data transfer in a network
- C. The time it takes for data to travel from source to destination
- D. The number of devices in a network

25. What is the significance of the TTL (Time-To-Live) field in an IP packet?

- A. It specifies the maximum file size that can be transmitted.
- B. It indicates the time duration of the data transfer.
- C. It limits the number of hops a packet can take in the network.
- D. It determines the encryption level of the packet.

26. Which two statements about the purpose of the OSI model are accurate? (Choose two)

- A. Defines the network functions that occur at each layer
- B. Facilitates an understanding of how information travels throughout a network
- C. Changes in one layer do not impact other layer
- D. Ensures reliable data delivery through its layered approach

27. Which of the following are related to the purposes of encapsulation and decapsulation? (Choose all that apply)

- A. Faster communication
- B. Interaction between different networks
- C. Layering of communication protocols
- D. Shorter length of datagrams.

28. Which of the following are specific network protocols? (Choose all that apply)

- A. HTTP      B. FTP      C. OSI      D. ISO      E. TCP      F. IP      G. TCP/IP

29. In the TCP/IP model, at which layer is the data unit referred to as a "frame"? (Choose one)

- A. Layer 1              B. Layer 2              C. Layer 3              D. Layer 4              E. Layer 5

30. What is the destination MAC address of a broadcast frame?

A. 00:00:0c:07:ac:01

B. ff:ff:ff:ff:ff:ff

C. 43:2e:08:00:00:0c

D. 00:00:0c:43:2e:08

E. 00:00:0c:ff:ff:f

31. Which of the following networks is most reliable? (Choose one)

A. Star network

B. Bus network

C. Ring network

D. Tree network

E. Fully meshed network

32. Into which geographical categories can networks be divided? (Choose all that apply)

A. Local Area Network

B. Ethernet

C. Internet

D. Wide Area Network

33. LAN is typically limited to a range of how many kilometers? (Choose one)

A. 0.1 km

B. 1 km

C. 10 km

D. 100 km

E. 1000 km

34. In a global positioning system (GPS), which method does the satellite use to communicate with the GPS receiver on the ground? (Choose one)

- A. Simplex communication
- B. Half-duplex communication
- C. Full-duplex communication.

35. What is the purpose of ICMP messages?

- A. to inform routers about network topology changes
- B. to ensure the delivery of an IP packet
- C. to provide feedback of IP packet transmissions
- D. to monitor the process of a domain name to IP address resolution

36. Which network service automatically assigns IP addresses to devices on the network?

- A. DHCP
- B. Telnet
- C. DNS
- D. traceroute

37. Which TCP segments will not be used in TCP three-way handshake? (Choose all that apply)

- A. SYN
- B. SYN+ACK
- C. ACK
- D. FIN
- E. FIN+ACK

38. What process involves placing one PDU inside of another PDU?

- A. encapsulation
- B. encoding
- C. segmentation
- D. flow control

39. When a TCP module sends a TCP segment, which layer will subsequently process the TCP segment to be sent? (Choose one)

- A. Application layer
- B. Transport layer
- C. Network layer
- D. Data link layer

40. A TCP host would indicate that it successfully received the packet with TCP sequence #800 and the 100 bytes of payload it was carrying, by sending ack# \_\_\_\_\_ back to the sender?

- A. 800
- B. 801
- C. 900
- D. 901

41. What “mechanism” does TCP use to control the amount of traffic it receives?

- A. Acknowledgements
- B. Window size announcements
- C. Sequence numbering
- D. Flags

42. A packet with sIP = 0.0.0.0 and dIP = 255.255.255.255 would most likely be associated with...

- A. an ARP request.
- B. a DHCP request.
- C. a DNS query.
- D. a directed broadcast packet.

43. The ARP protocol is given a/an \_\_\_\_\_ address and—hopefully—will return a/an \_\_\_\_\_ address.

A. IP ->MAC

B. IP -> name

C. MAC -> IP

D. name -> IP

44. A computer that is continually “listening” on port 25, is most likely an...

A. smtp (mail) server.

B. ftp (file) client.

C. ftp (file) server.

D. smtp (mail) client.

45. What is the correct order of data encapsulation while one goes *downwards* from the top to bottom of our hybrid five layer model?

A. Data, Frame, Packet, Datagram, Bit

B. Data, Frame, Datagram, Packet, Bit

C. Data, Packet, Frame, Datagram, Bit

D. Data, Datagram, Packet, Frame, Bit

E. Data, Packet, Datagram, Frame, Bit

F. Data, Datagram, Frame, Packet, Bit

46. Why is port number included in the TCP or UDP header of a datagram?

A. To indicate the correct router interface that should be used to forward a datagram

B. To identify which switch ports should receive or forward the datagram

C. To determine which Layer 3 protocol should be used to encapsulate the data

D. To enable a receiving host to forward the data to the appropriate application

E. To allow the receiving host to assemble the packet in the proper order

47. Which one of the following is the purpose of the TCP three-way handshake?
- A. Sending echo requests from the source to the destination host to establish the presence of the destination.
  - B. Determining the IP address of the destination host in preparation for data transfer
  - C. Requesting the destination to transfer a binary file to the source
  - D. Synchronizing window size and sequence numbers between source and destination in preparation for data transfer
  - E. None of the above
48. Which features characterize the UDP transport layer protocol? (Choose Two)
- A. Packet acknowledgment may reduce overhead traffic
  - B. Duplication control ensures error free delivery
  - C. Packets are not acknowledged which reduces overhead traffic and enhance performance.
  - D. Requires handshaking before transmission.
  - E. Encapsulate data into datagram to ensure end to end data integrity
49. Packets in one TCP connection can be distinguished from packets in another TCP connection by:
- A. Source port, destination port, protocol number
  - B. Source IP address, destination IP address, protocol number
  - C. Source port, destination port, source IP address, destination IP address
  - D. Source IP address, destination IP address
  - E. Source port, destination port
  - F. Protocol number
50. A network administrator uses the ping command to check for points of failure in the network. Which protocols will be used during this process? (Choose two)
- A) ICMP                      B) TCP                      C) ARP                      D) UDP
51. SMTP is most commonly used for:
- A) Managing devices (such as routers and switches) on a network
  - B) Sending emails from an email client to an email server
  - C) Retrieving emails from an email server to an email client
  - D) Mapping IP addresses to hardware addresses
52. Which transport protocol allows real-time applications (e.g. video streaming) to send packets without waiting for ACKs?
- A) TCP                      B) DHCP                      C) IP                      D) ARQ                      E) UDP

53. What is the port number of POP3?

- A) 110                      B) 90                      C) 80                      D) 49

54. Who originally designed TCP/IP

- A) The Department of Defense      B) Novell      C) IBM      D) Xerox

55. UDP is:

- A) Not a part of the TCP/IP suite                      B) Connection orientated and reliable  
C) Connection oriented and unreliable                      D) Connectionless and unreliable

56. Where is a hub specified in the OSI model?

- A) Session layer      B) Physical layer      C) Data Link layer      D) Application layer

57. What are two characteristics shared by TCP and UDP? (Choose two.)

- (a) default window size                      (d) 3-way handshake  
(b) connectionless communication      (e) ability to carry digitized voice  
(c) port numbering                      (f) use of checksum

58. The purpose of the Protocol Number field in the IP header is:

- A) Indicate the lower layer (data link) protocol that the data is intended for  
B) Indicate the protocol options used by IP  
C) Indicate the next higher layer protocol that the data is intended for  
D) Indicate the version of IP being used

59. What is the purpose of flow control?

- (a) To ensure that data is retransmitted if an acknowledgment is not received.  
(b) To reassemble segments in the correct order at the destination device.  
(c) To provide a means for the receiver to govern the amount of data sent by the sender.  
(d) To regulate the size of each segment.

60. Which Protocol does DHCP use at the Transport layer?

A) IP                      B) TCP                      C) UDP                      D) ARP

61. What is the primary purpose of the routing process?

(a) To map IP addresses to MAC addresses      (c) To switch traffic to all available interfaces

(b) To propagate broadcast message      (d) To find paths from one network or subnet to another

62. Error detection at the data link level is achieved by

(a) Bit Stuffing      (b) Cyclic Redundancy Code      (c) Hamming Code      (d) Equalization

63. Which value that is contained in an IPv4 header field, is decremented by each router that receives a packet?

(a) Header Length      (b) Differentiated services      (c) Time-to-live      (d) Fragment Offset

64. Start and stop bits are used in serial communication for

(a) Error detection (b) Error correction (c) Synchronization (d) Slowing down the communication

65. The minimum length of data in Ethernet frame is

(a) 64 Bytes                      (b) 46 Bytes                      (c) 1500 Bytes                      (d) 1518 Bytes

66. Which IPv4 class of addresses provides the most networks

(a) Class A                      (b) Class B                      (c) Class C                      (d) Class D                      (e) Class E

67. Which IPv4 class provides the highest number of host addresses per network?

(a) Class A                      (b) Class B                      (c) Class C                      (d) Class D                      (e) Class E

68. Your company has a LAN in its downtown office and has now set up a LAN in the manufacturing plant in the suburbs. To enable everyone to share data and resources between the two LANs, what type of device(s) are needed to connect them? Choose the most correct answer.

(a) Modem                      (b) Cable                      (c) Hub                      (d) Router

69. What are two functions that are provided by the network layer? (Choose two.)
- (a) directing data packets to destination hosts on other networks
  - (b) placing data on the network medium
  - (c) carrying data between processes that are running on source and destination hosts
  - (d) providing dedicated end-to-end connections
  - (e) providing end devices with a unique network identifier
70. Which of the following summation operations is performed on the bits to check an error-detecting code?
- (a) Codec
  - (b) Coder-decoder
  - (c) Checksum
  - (d) Attenuation
71. An error-detecting code inserted as a field in a block of data to be transmitted is known as
- (a) Frame check sequence
  - (b) Error detecting code
  - (c) Checksum
  - (d) flow control
72. Which of the following is the address of the router?
- (a) The IP address
  - (b) The TCP address
  - (c) The subnet mask
  - (d) The default gateway
73. What is the first octet range for a class C IP address?
- (a) 192 – 255
  - (b) 192 – 223
  - (c) 192 – 226
  - (d) 128 – 191
  - (e) 1 - 126
74. Ethernet networks can be cabled in a number of topologies, depending on what works best in each environment. As more nodes are added, the efficiency of Ethernet decreases. Select the best answer as to why Ethernet becomes less efficient as size increases.
- (a) Network collisions occur
  - (b) Repeaters cannot increase the signal strength sufficiently
  - (c) Cable terminators do not reflect the signal properly
  - (d) Cable terminators do not absorb the signal properly
  - (e) "Line echo" occurs due to the impedance of the cable
75. The IEEE 802 project of the 1980s involved further defining the lower two layers of the OSI model. A number of standards were agreed upon during that time. Which of the following is the standard for Ethernet?
- (a) 802.2
  - (b) 802.3
  - (c) 802.4
  - (d) 802.5
  - (e) 802.6

76. A medium access control technique for multiple access transmission media is
- (a) Aloha                      (b) Amplitude                      (c) Angle modulation                      (d) Attenuation
77. Which of the following MAC protocols is used by Ethernet?
- (a) Slotted Aloha      (b) CSMA/CA                      (c) CSMA/CD                      (d) TDMA                      (e) FDMA
78. Which subnet would include the address 192.168.1.96 as a usable host address?
- (a) 192.168.1.64/26                      (b) 192.168.1.32/27                      (c) 192.168.1.32/28                      (d) 192.168.1.64/29
79. IEEE project 802 divides the data link layer into an upper \_\_\_\_\_ sub-layer and a lower \_\_\_\_\_ sub-layer
- (a) HDLC, PDU                      (b) PDU, HDLC                      (c) MAC, LLC                      (d) LLC, MAC
80. On a VLSM network, which mask should you use on point-to-point WAN links in order to reduce the waste of IP addresses?
- (a) /27                      (b) /28                      (c) /29                      (d) /30
81. What is the difference between an Ethernet hub and an Ethernet switch?
- (a) A switch can be used in a star topology; a hub cannot be used in a star topology.
- (b) A switch forwards frames to other stations; a hub does not forward frames.
- (c) A switch is used to send IP datagrams; a hub does not send IP datagrams.
- (d) A hub always sends a frame to all stations; a switch may send a frame to a specific station.
82. Consider Stop-and-Wait, Go-Back-N and Selective-Reject ARQ protocols. Which statement is true?
- (a) Go-Back-N will always be more efficient than Stop-and-Wait and Selective-Reject.
- (b) Selective-Reject requires the receiver to buffer out-of-order frames
- (c) In Go-Back-N the sender may send multiple frames before waiting for an ACK; in Selective-Reject the sender must wait for an ACK after sending a single frame.
- (d) Selective-Reject uses sequences numbers, whereas Go-Back-N and Stop-and-Wait do not.
- (e) In Stop-and-Wait, ACK frames are never lost.
- (f) Stop-and-Wait does not require timers to be used.

83. Which is the most common data rate used today in IEEE 802.3 Ethernet LANs?

- (a) 1 Mb/s    (b) 10 Mb/s    (c) 54 Mb/s    (d) 100 Mb/s    (e) 1000 Mb/s

84. A disadvantage of circuit switching (versus datagram packet switching) is:

- (a) It is difficult to guarantee performance for delivery of data
- (b) As new connections are requested they may be blocked
- (c) Data is likely to be dropped at switches
- (d) Switches are complex because they must process every piece of data
- (e) A larger delay (than datagram packet switching) will always occur when transferring the same amount of data

85. Compared to other common topologies in LANs (such as bus, mesh, ring), which of the following is a disadvantage of using a star topology for LAN?

- A) Failure of a special node results in no stations being able to communicate
- B) Poor performance due to sharing of point-to-multipoint links between users
- C) Each station requires many network interfaces
- D) Failure of a single link results in no stations being able to communicate

86. Normally TCP would be implemented as:

- A) Software in the operating system
- B) An application installed by users that want to use UDP
- C) Part of a user application, such as web browser or email
- D) Hardware on the LAN/WAN interface cards
- E) Device drivers that control the LAN/WAN interface cards

87. Which protocol provides flow control, error control and connection management:

- A) DNS            B) ICMP            C) ARP            D) TCP            E) UDP

88. Which protocol provides error reporting capabilities in the Internet:

- A) TCP            B) ICMP            C) IP            D) ARP            E) UDP

89. What are the three measures of effective data communications?

- A) Accuracy, complexity, cost
- B) Delivery, complexity, timeliness
- C) Accuracy, delivery, timeliness
- D) Bandwidth, delivery, timeliness

90. A model for communication systems consists of the following components (the list must be in the correct order):

- A) Source → Transmitter → Receiver → Destination
- B) Source → Transmitter → Receiver → Destination → Transmission System
- C) Source → Transmitter → Transmission System → Receiver → Destination
- D) Transmitter → Source → Transmission System → Destination → Receiver
- E) Transmission System → Source → Transmitter → Receiver → Destination

91. Which protocol maps domain specific addresses into logical addresses

- A) DNS
- B) ICMP
- C) IP
- D) ARP
- E) UDP

92. The standard suit of protocols used by the Internet, intranets, extranets, and some other networks

- A) TCP/IP
- B) Protocol
- C) Open Systems
- D) Internetwork processor
- E) Hybrid hub

93. Information systems with common hardware, software, and network standards that provide easy access for end users and networked computer systems, Select the best fit for answer.

- A) TCP/IP
- B) Protocol
- C) Open Systems
- D) Internetwork processor

94. Layer 1 of the OSI model is

- A) Physical layer
- B) link layer
- C) transport layer
- D) network layer

95. What is the standard protocol for network management features?

- A) SNA
- B) SMTP
- C) SNMP
- D) SMS

96. Which of the following TCP/IP protocol is used for remote terminal connection service?

- A) TELNET
- B) FTP
- C) RARP
- D) UDP

97. What is the main reason the OSI model was created.

- A) To create a layered model larger than the DoD model.
- B) So application developers can change only one layer's protocols at a time.
- C) So different networks could communicate.
- D) So Cisco could use the model.

98. The type of switching used in IP networks is:

- A) Circuit switching
- B) Datagram packet switching
- B) Ethernet switching
- D) Virtual Circuit Packet Switching

99. Select the correct set of features provided by IP:

- A) Addressing, error control, security
- B) Connection-less, addressing, data delivery
- C) Addressing, flow control, quality of service
- D) Connection-oriented, data delivery, flow control
- E) Flow control, addressing, data delivery
- F) Addressing, error control, connection-less

100. The process of finding a path through a network is called:

- A) congestion control
- B) forwarding
- C) fragmentation
- D) routing